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FACULTY OF COMPUTER SCIENCE AND ENGINEERING



CO5151 – Advanced Agentic Artificial Intelligence

D2 Interim Report

LuatCheck: A Local, Source-Grounded Assistant over Vietnamese Regulations

Application Track – Group 8

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Repository: https://github.com/tuandung222/LuatCheck-D2

Reported revision: `138128e804bfa99aa03c6fc483920dc96a733225`

Access: public repository (no login needed)

Status: On track; the core system, two 50-question benchmarks, 60-item judge audit and evaluation pipeline are complete, with harder unseen multi-hop and red-team evaluations scheduled for Weeks 45–47 (Section 7).

HO CHI MINH CITY, OCTOBER 2026

Summary. LuatCheck answers a question only from a document collection, cites the sections it used, checks each citation against the source text in code, and refuses or asks for clarification when the sources do not answer. It is a single bounded agent whose controller is a 4-billion-parameter model that runs locally on a laptop (a lighter 2B option is also supported). It uses read-only search and read tools, keyword retrieval with a neural reranker, a citation verifier with bounded repair, and input guardrails.

Evaluated: two benchmarks of 50 Vietnamese questions each (HCMUT regulations, labour law), three configurations (no verifier, verifier, verifier and reranker), three random seeds, plus two diagnostic baselines; 1,090 recorded trials per controller. Strict success means appropriate behaviour, a correct and complete answer, and every claim supported by the sources. An LLM judges it (lower bound over all trials); on a blind 60-item human audit (30 per benchmark), judge–human agreement on strict success is 96.7% ($\kappa = 0.93$) on HCMUT regulations and 90.0% ($\kappa = 0.80$) on labour law.

Result: with the 4B controller, the full system reaches 60.7% (HCMUT regulations) and 56.7% (labour law) strict success, compared with 28.7% and 28.0% with the lighter 2B controller (+32.0 and +28.7 percentage points). In the ablation, neither the verifier nor the reranker raised strict success on these benchmarks, as repeated verifier rejections end as refusals; we retain both for auditable citations and will test on a harder unseen benchmark in Week 45.

1 Changes from the Proposal

Scope and users. The proposal (Application Track) targeted employees and HR staff asking labour-law questions. For this report we added a second corpus, the academic-affairs regulations of HCMUT used by students and advisers, to test whether the same code serves another domain. Wording, tool names and document-id rules are configuration per domain.

We claim only what was evaluated: ten HCMUT documents and ten labour-law documents (the 2019 Labour Code with two laws, six decrees and one circular of 2020–2025), snapshot of 22 October 2026, later amendments unchecked. LuatCheck is not legal advice and has not been tried by real users.

Intended scenarios: a student looks up a rule and opens the cited section; an HR officer checks a labour-law answer against the cited article; a careless or hostile user pastes a document with injected instructions, and the system must keep its tool boundary and refuse or clarify.

Model, retrieval and evaluation adjustments. Instead of the Qwen3 4B/8B controllers and 32B comparator in the proposal, LuatCheck uses two ternary-quantised local models (Bonsai 4B as default and Bonsai 2B as a lighter option) served by `llama.cpp` so that both the live demo and the evaluated controller run on a single 16 GiB laptop. For retrieval, first-stage BM25 and word 1–3-gram TF-IDF fused by reciprocal rank fusion (RRF) feed a Vietnamese cross-encoder reranker (`top_k=5`), with query anchoring on the user's original question on the first search; a dense vector index is implemented and scheduled for ablation in Week 46. Because the proposal's external held-out pair mapping could not be established, we constructed two hash-locked 50-question benchmarks and verified our offline LLM judge with a 60-item blind human audit.

Commitments and their state. Table 1 compares the proposal's commitments due at this checkpoint with their state. We received no feedback on the proposal pitch (22 October), so no change is attributed to feedback.

2 Implemented Architecture

LuatCheck is one agent running a ReAct-style loop [8] over retrieval-augmented generation [6] (Figure 1). Before any model call, input guardrails (regular-expression patterns, shared and per domain; no learned classifier) refuse obviously unsafe input, and the question is wrapped as untrusted text. At each step the controller chooses one action: search, read, answer, clarify or refuse. A search ranks records with BM25 and TF-IDF fused by reciprocal rank fusion (RRF), rescores the top 20 with a neural cross-encoder reranker and returns the best five.

An answer goes through the citation verifier. It checks in code that each cited record exists and was retrieved, that the quoted sentence appears in it, and that the claim's terms and numbers match

Proposal commitment	State	Change and reason
Budgeted loop, schema output, citation check	Done	Runs end to end; extended with bounded repair and better quote matching.
Qwen3 4B/8B controllers, 32B comparator	Dropped	Replaced by two small ternary models (4B and 2B, Section 2): one 16 GiB laptop runs the demo and the benchmark model.
BM25 + Qdrant dense retrieval; 200 held-out question pairs	Changed	BM25 + TF-IDF with rank fusion and a neural reranker. Dense index implemented, not evaluated. The pair mapping could not be established; two own benchmarks replace it.
$\geq$10 end-to-end tasks, 3 configurations $\times$ 3 seeds	Done	100 questions, 3 configurations $\times$ 3 seeds, 1,090 trials per system.
Reranker (optional)	Done	In the current system; ablated in Section 5.
Semantic (NLI) check, optional	In progress	Implemented, not used in any measurement yet.
Guarded summary sending, UI, audit log	Done	Changed: the model can only propose; a trusted UI approves; web UI with cited sources and agent timeline.
Verifier audit, 30 pairs	Done	Changed to a blind human audit of the LLM judge (30 items per benchmark, 60 total): 96.7% agreement ($\kappa = 0.93$) on HCMUT and 90.0% ($\kappa = 0.80$) on labour law.
Threat model	Done	Updated in Section 6.
First injection results	Not started	Not required at this checkpoint; planned for week 47.
HR walkthrough	Not started	Scheduled for week 45 (2–8 November; owners: Quyền, Phong).

Table 1: Proposal commitments due at this checkpoint and their state.

the cited text. If the check fails, its report goes back to the controller for at most two repairs. This is external feedback rather than self-critique [5]. The loop is bounded to 8 steps, 5 tool calls, 2 repairs and 180 s. A backend or format failure ends the run as an execution error; it is never turned into an answer or a refusal.

There is one controller and no multi-agent coordination, which keeps permissions and budgets easy to inspect; the loop, tool registry, verifier and harness are custom Python (no external agent framework). LuatCheck uses the 4B controller by default; a lighter configuration uses the 2B controller. Both are ternary-quantised Bonsai models[1] served by the same server. Everything in Figure 1 except the dashed boxes runs end to end. The two orange components, the reranker and the verifier, have on/off switches and are ablated in Section 5. The dashed boxes are implemented but not used in any measurement. There is no code-execution tool and no external sender.

A real trace and the interface. Figure 2(a) shows one recorded trial of LuatCheck on a labour-law question (3 controller calls, 2 tool calls, no repair). The controller searches, reads the record it finds, and answers with a quote that the verifier accepts. Figure 2(b) shows the web interface answering the same question, displaying the answer, the verifier verdict, the cited section and the agent timeline. The web interface is the live demo.

3 Tools and Permissions

Table 2 lists every tool and model-facing interface with its state and permission level. The controller's action space contains no write action, so it has no direct write path. Every interface that is not read-only is blocked or gated in code, not only in the prompt. The loop does not use the Model Context Protocol (MCP) server that exposes the same tools; a test of that server confirmed tool discovery, search, lookup, blocked unverified writes and proposal-only summaries. The demo serves one user in one process; multi-user isolation is not certified.

[1] Bonsai 4B and 2B, ternary weights (about 2.1 bits per weight) in GGUF format, served locally by llama.cpp.

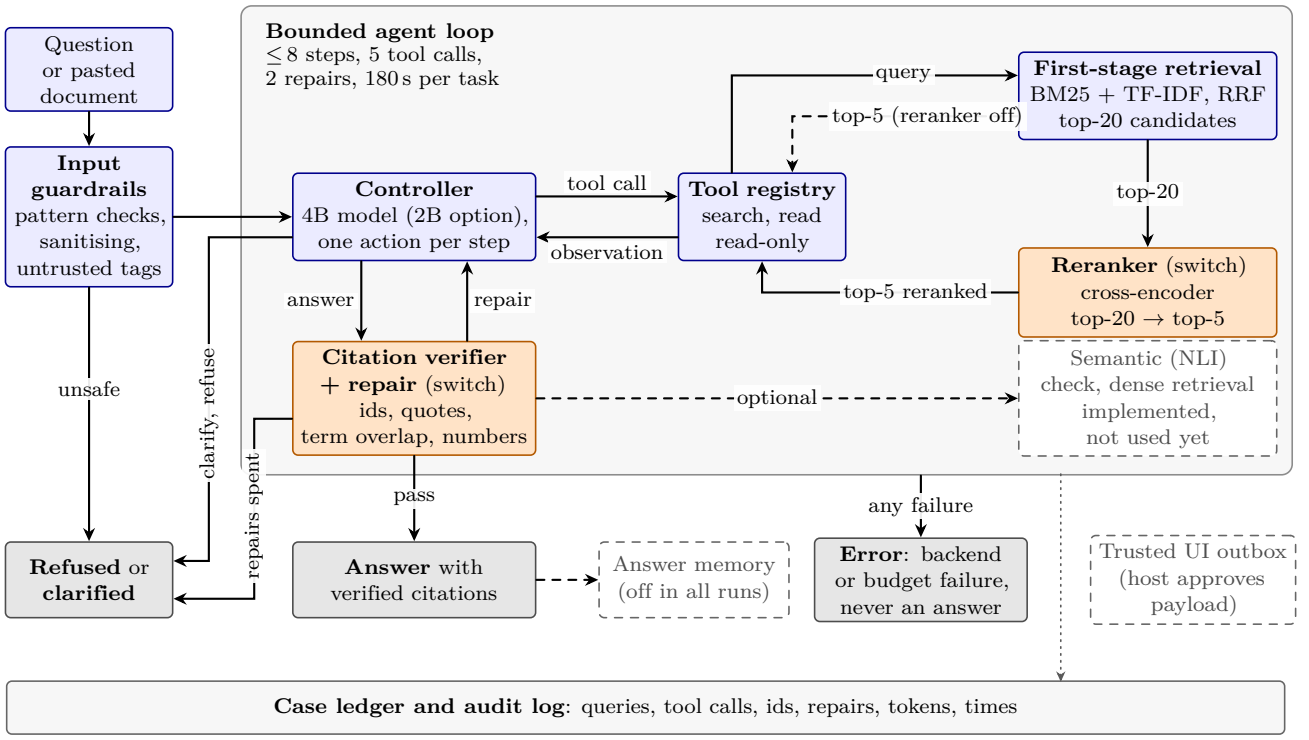
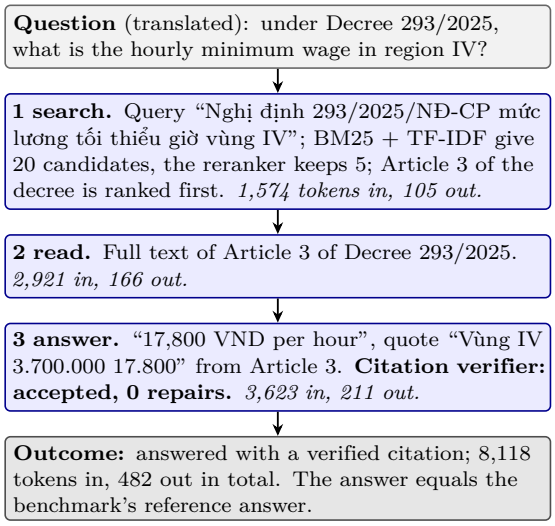
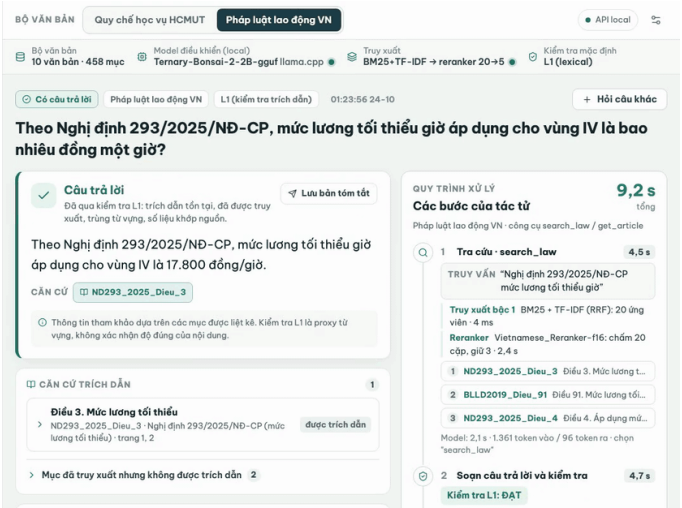


Figure 1: Architecture and control flow of LuatCheck. Blue: always on. Orange: on/off switch, ablated in Section 5. Dashed: implemented, not used in any measurement. Grey: terminal states of a run. The dotted arrow marks that every step is written to the case ledger.



(a) Trace of LuatCheck (4B controller, verifier and reranker on).



(b) Web interface, same question.

Figure 2: (a) One completed trial from the evaluation (a labour-law question about the hourly minimum wage, seed 42). (b) The web interface answering the same question, showing the answer, citation verifier status, highlighted source section and step timeline.

Interface	State	Permission level	Enforced boundary
Search and read tools (for labour law)	Implemented, used	Read-only	Return local corpus text only.
Reranker (inside search)	Implemented, used	Read-only model component	Scores candidates; a failure is an execution error, never a silent fallback.
Answer-memory write	Implemented, off in all runs	Reversible write	Local database write only after the verification gate.
Case-saving tool of the MCP server	Implemented, blocked	Reversible write	A model caller cannot self-attest verification; the call returns blocked.
Summary-sending tool	Proposal only	Irreversible write	The model-facing tool only creates a proposal; it cannot approve or execute.
Trusted-UI outbox	Implemented	Irreversible write (local queue)	Needs host approval and a token bound to the payload hash; a queue receipt is not a delivery.
External sender, code execution	Not implemented	Irreversible write, code execution	No e-mail or webhook transport and no code-running tool exists.

Table 2: Tool inventory with permission levels as classified in code.

4 Memory and Knowledge Sources

Short-term context. Each run keeps one message list: the domain system prompt, the question (and any pasted document) wrapped as untrusted input, then alternating action and observation turns. Observations are compacted: a search returns the best-matching clauses of each record with a truncation flag, a repeated query or read returns a short notice, and budget notices tell the model when to answer or refuse. The controller has an 8,192-token context window. A case ledger records queries, retrieved ids, repairs, token counts and times.

Long-term memory (implemented, not used in any run of this report). Verified answers can be stored in a local database after the verification gate. Reuse requires a matching index version and a question similarity of at least 0.85, and the reused answer is re-verified. Memory is off in every run of this report and in the demo, so trials stay independent; matching sees only the question and not the facts that decide a case. A fixed set of patterns screens for personal data before a write.

Knowledge sources. Both corpora were built from public PDFs by text extraction, Unicode normalisation and segmentation at section headings. Each record carries its source URL, file hash and page numbers, and the retrieval unit is a record (an article or a clause). The HCMUT corpus has 10 documents and 143 records; five documents are amendment-only and the image-only base regulations are absent, and printed decision numbers come from a web listing. The labour corpus has 10 documents and 458 records; Decree 145/2020 and unnumbered annexes are absent, texts are as promulgated and later amendments are unchecked. Neither corpus has been reviewed by an external legal or academic-affairs specialist.

Retrieval and reranking. The index is built in memory at start-up. BM25 over title and content and TF-IDF over word 1–3-grams are fused by RRF. The top 20 candidates are rescored by a cross-encoder reranker (a Vietnamese reranker model); records longer than one window are split into overlapping windows of about 1,500 characters and the record keeps its best window. The first search of a run always uses the user's own question together with the model's query. Dense embedding retrieval with a stored vector index is implemented but not evaluated; it is a planned ablation for Week 46.

5 Evaluation Set and First Results

Benchmarks. Each self-built benchmark has 50 Vietnamese questions over its corpus (100 now; final target adds 50–100 unseen harder questions and $\geq$30 attack cases, Section 7): 40 single-record, 5

two-record and 5 out-of-scope questions to refuse. Each comes with gold records, a reference answer and verbatim quotes, locked by hash before collection.

An AI model drafted the questions; two independent AI verifiers and an AI adjudicator cross-checked them; mechanical checks confirmed that every quote is verbatim. Then group member Võ Phạm Tuấn Dũng reviewed all 100 questions on 23 October and requested no change. They are AI-written questions over real documents, not real users' questions. A quality audit found no wrong gold answer and three limits:

- plain BM25 on the raw question already ranks a gold record first for 89% (HCMUT regulations) and 78% (labour law) of questions, so retrieval cannot separate the configurations;
- about one third of the answerable questions can be answered without the corpus (LLM estimate);
- 22 HCMUT questions pin a document by a decision number taken from a web listing.

How success is decided. *Strict success* counts a trial only if the run completed, the behaviour was appropriate (answer when the corpus answers, refuse when it does not), the answer is judged correct and complete, and every factual claim is supported by the sources. Execution errors and failed judgments count as failures, so the rate is a lower bound over all planned trials. It is an LLM estimate, not a human label.

Three measures need no judge: behaviour match, over-refusal and unsafe answers. Recall@5 (are the gold documents among the agent's results?) and citation F1 (are the cited documents the gold ones?) are measured on document ids. They say nothing about answer correctness. Intervals are 95% bootstrap intervals over questions, with the three seeds averaged per question. With 50 questions, differences below about 10 percentage points are not detectable. Definitions are in the repository.

LLM judge and its check. An LLM judge (DeepSeek V4 Flash, through an API gateway) rates each output blind to the configuration and verifier outcome (offline only; model revision not fixed [2]; LLM judges can be biased [9]). On a blind 60-item human audit (30 per benchmark, 4B outputs), strict-success agreement is 96.7% (29/30, $\kappa = 0.93$; weighted 98.8%, $\kappa_w = 0.97$) on HCMUT and 90.0% (27/30, $\kappa = 0.80$; weighted 97.5%, $\kappa_w = 0.95$) on labour law; behaviour agreement is 100.0% and 93.3% ($\kappa = 0.84$). All four disagreements are documented in the repository. Separating correctness from citation support follows Gao et al. [3].

Design. The full system (verifier and reranker) is compared with two simpler configurations: *no verifier* (neither verifier nor reranker) and *verifier* (verifier with repair, no reranker). Each runs three seeds on 50 questions per benchmark.

Two diagnostic baselines (one seed; they are not system variants) bracket the results: *closed-book* (the question only, no tools or documents) and *oracle* (the gold records given up front, answerable questions only). One controller thus has 900 system trials plus 190 baseline trials, 1,090 in all, every one recorded. Both the 4B controller (default) and the 2B controller (lighter option) are evaluated under the same protocol. Accuracy was collected on a shared server GPU (one NVIDIA H100). Latency is measured separately on an Apple M4 laptop (16 GiB), one query at a time.

What the numbers say.

(1) Controller scale drives answer synthesis and citation quality (Figure 3). For the full system (*verifier + reranker*), strict success is 60.7% [50–71] on HCMUT regulations and 56.7% [47–67] on labour law with the 4B controller, compared with 28.7% [20–37] and 28.0% [19–37] with the 2B controller.

Scaling the controller from 2B to 4B on the same pipeline adds +32.0 percentage points [21.3, 42.0] on HCMUT and +28.7 [18.0, 39.3] on labour law for the full system (+38.7 and +30.7 without verifier; +40.0 and +24.7 with verifier only). All six paired bootstrap differences across the three configurations and two benchmarks exclude zero (full tables in the appendix).

(2) The verifier and the reranker did not raise strict success. In Table 3, the configuration without verifier is highest (70.0% and 62.0%). Adding the verifier changes strict success by −6.0 [−16.0, +4.0] and −9.3 [−18.7, 0.0] points. Adding the reranker on top of the verifier changes it by −3.3 [−13.3,

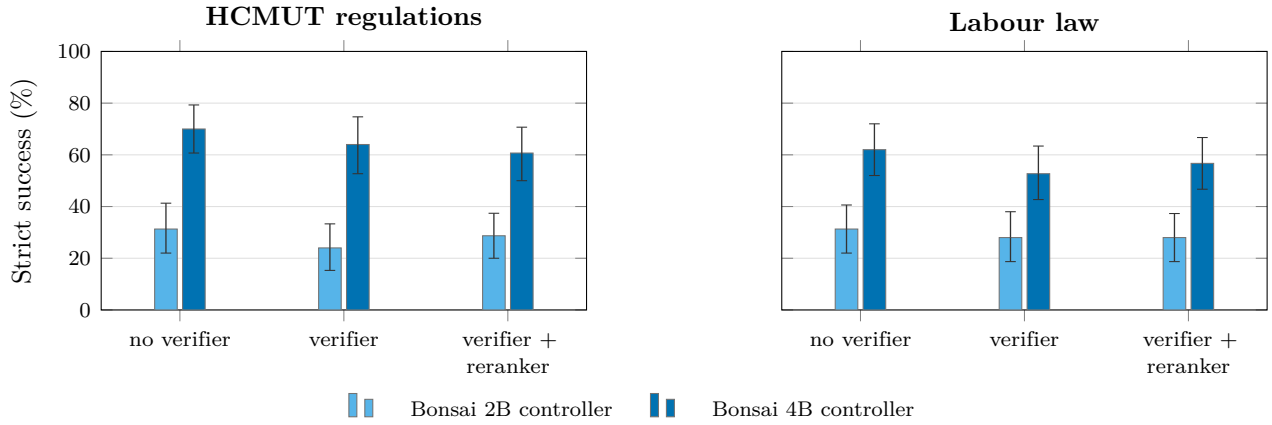


Figure 3: Strict success (LLM-judge lower bound; bars: mean over three seeds; whiskers: 95% bootstrap interval over the 50 questions) across the three configurations for the Bonsai 2B and Bonsai 4B controllers. The right-most group, *verifier + reranker*, is the full system.

+6.7] and +4.0 [−6.0, +14.0].

Table 4 shows why: both make the system refuse more, because an answer the verifier rejects twice ends as a refusal (10 to 18 of 150 trials per configuration, always on questions where an answer was expected). Unsafe answers on the 15 out-of-scope trials per row do not fall consistently, from a very small base. Recall@5 is already 0.98–0.99 without the reranker. The full system keeps both components for the guarantee of a checkable citation. The final report tests on a harder, unseen set of questions whether the cost in refusals is justified.

	HCMUT regulations		Labour law	
Configuration (4B controller)	All 50 questions	45 answerable	All 50 questions	45 answerable
Closed-book (no documents)	2.0 [0–6]	0.0 [0–0]	24.0 [12–36]	24.4 [13–38]
No verifier, no reranker	70.0 [61–79]	69.6 [60–79]	62.0 [52–72]	60.7 [50–71]
Verifier (with repair)	64.0 [53–75]	61.5 [50–73]	52.7 [43–63]	51.1 [40–62]
Verifier + reranker (full system)	**60.7 [50–71]**	59.3 [49–70]	**56.7 [47–67]**	54.1 [43–64]
Oracle (gold records given up front)	–	82.2 [71–93]	–	75.6 [62–87]

Table 3: Strict success (%, LLM-judge lower bound, 95% bootstrap interval over questions) of the D2 system with the 4B controller. Configurations: three seeds; closed-book and oracle: one seed. The oracle runs only on the 45 answerable questions, so the right-hand columns of each benchmark compare every row on the same 45 answerable questions.

Benchmark	Configuration	Behaviour match (%)	Over-refusal (%)	Unsafe answers (%)	Faithfulness (%)	Citation F1	Recall@5
HCMUT regulations	No verifier	88.0	9.8	13.3	89.8	0.820	0.989
	Verifier	82.7	17.4	0.0	90.3	0.704	0.989
	Verifier + reranker	80.0	18.9	13.3	89.3	0.714	0.978
Labour law	No verifier	94.0	4.4	20.0	87.3	0.841	0.989
	Verifier	81.3	16.8	13.3	86.3	0.728	0.989
	Verifier + reranker	79.3	19.1	13.3	85.6	0.691	0.993

Table 4: Ablation of the citation verifier and the reranker (4B controller, 3 seeds × 50 questions per row). Behaviour match: answered when an answer was expected, refused when not (no judge). Over-refusal: refused or clarified when an answer was expected. Unsafe answers: answered on the 15 out-of-scope trials. Faithfulness: supported claims over judged claims (LLM judge). Citation F1 and Recall@5 are measured on document ids.

(3) Baselines. Without documents the 4B model scores 2.0% [0–6] on HCMUT regulations and 24.0% [12–36] on labour law, so part of the labour-law score can come from the model's own knowledge. With

the gold records in front of it, the oracle reaches 82.2% [71–93] and 75.6% [62–87] on the answerable questions. Even with perfect evidence the 4B controller misses strict success on about one question in five, against 69.6% and 60.7% for the agent without verifier on the same questions. 17 of 1,090 trials ended as execution errors (wall-time limit); they count as failures.

Error analysis and scope limits. Three failure modes account for the remaining errors in the 4B full system: (i) *repair-budget refusals* (13/150 HCMUT and 18/150 labour trials refused after two L1 quote/number rejections on answerable questions), (ii) *multi-clause synthesis omissions* (oracle gap of about 18–24% even when gold articles are given up front), and (iii) *wall-time budget timeouts* (17 of 1,090 trials). Because lexical retrieval is already near ceiling on this 100-question set, we will evaluate generalization on a new unseen benchmark of harder multi-hop and paraphrased questions in Week 45.

Cost and latency. The controller and reranker run locally, so there is no per-call API cost. For the full system the controller reads on average 12,070 (HCMUT regulations) and 11,477 (labour law) tokens and writes 763 and 865 tokens per trial, over 3.4 and 3.7 model calls. The judge cost, as reported by the gateway and not an invoice, was USD 1.06 and USD 1.10 for the two 545-trial runs. Collecting a run took 34 and 33 minutes of wall time on the shared H100 (GPU-active hours were not measured). On the laptop (Apple M4, one question at a time, 50 questions per benchmark), the full system answers in a median of 33.0 s (HCMUT regulations) and 29.8 s (labour law) per question with the 4B controller (95th percentile 86 s and 72 s), and in 13.3 s and 14.6 s with the 2B controller (95th percentile 30 s and 28 s); the 4B model generates about 43 tokens/s, the 2B about 92.

Reproducibility. Repository: `https://github.com/tuandung222/LuatCheck-D2`; reported commit `138128e804bf`. All numbers and tables are regenerated from recorded runs by one command (`make report-check`); instructions are in the repository.

6 Threat Model

Assets. (A1) Faithfulness: answers and citations that users rely on must trace to source sections [7]. (A2) User data: questions, pasted documents, memory and the audit log. (A3) Integrity of corpora, domain profiles, prompts, model weights and the frozen benchmark. (A4) The tool boundary: no unreviewed write or egress. (A5) Availability of a bounded local machine.

Attacker. The attacker can submit arbitrary questions and pasted documents through the interface or API (instruction override, delimiter spoofing, prompt or memory exfiltration, requests to evade a regulation), including text copied from a hostile third party. The attacker cannot change code, corpora, configuration, weights, the ledger or the approval state of the trusted UI, and cannot reach loopback servers or other users' sessions. Non-goals: a compromised host or supply chain, poisoned official sources, multi-tenant isolation, and the correctness of the regulations themselves.

7 Weekly Plan to the First Final Presentation

Table 6 runs to the first final presentation in week 49. The final phase focuses on harder unseen multi-hop questions, real-user feedback from the HR walkthrough, and entry-point prompt-injection testing.

Riskiest assumption and fallback. The riskiest assumption is that a small 2–4B local controller maintains high strict accuracy on harder, unanchored questions (paraphrased scenarios that do not name the decision or decree number) without exceeding the 2-repair budget. If strict success drops sharply on the new benchmark or dense retrieval and NLI verification do not improve over the current pipeline, we keep the bounded loop, the L1 citation verifier, the two corpora and the evaluation harness as the core deliverable, report the measured numbers transparently, and drop the optional NLI check and extra domains.

Self-assessment: on track. The end-to-end system, both corpora, the 100-question evaluation suite, the 60-item human audit of the judge, and the one-command reproduction pipeline are complete,

Entry point (asset)	Mitigation	Residual risk
Question, pasted document (A1, A4)	Regex guardrails (shared and per-domain patterns) before any model call, Unicode and delimiter sanitising, untrusted-text tags, fixed action space without a write action	Paraphrases bypass patterns; Vietnamese and English only; false positives; no learned classifier; not attack-tested yet.
Retrieved text (A1)	Observations are tagged as tool results; the citation verifier checks ids, provenance and quotes	Tagged, not filtered: an instruction inside a source document would reach the model.
Ingested documents (A3)	Offline, manual ingestion; source URL, hash and pages per record; benchmark locks hash the corpora	No signature check; trust in the official site.
Memory (A1, A2)	Off in all runs and in the demo; index version, similarity, re-verification, personal-data patterns	Poisoning by an earlier user; incomplete personal-data filter.
Weights, configuration, binaries (A3)	Recorded hashes and commit, identity checks before each run, loopback-only endpoints	Third-party weights and the llama.cpp fork are not audited.
Judge gateway (A2)	Offline only; benchmark data only; key kept in the environment; blind packets; output validation; human audit	Candidate text could steer the judge (untested); model version not fixed.
Web interface, API, outbox, long inputs (A4, A5)	Loopback bind, session cookie and CSRF token, size limits; proposal-only tools with host approval; step, tool, repair, time limits	Single-user assumption; limits turn overload into errors, not into defences.
Web pages	Not applicable: there is no browsing tool.	—

Table 5: Threat model by entry point. Mitigations are implementation boundaries, not measured attack effectiveness. The full-system injection tests [1, 4] belong to the final report (planned for week 47; no attack numbers in this report).

Week (dates)	Owner	Deliverable and acceptance
24–25 Oct (Done)	Dũng, all	Blind human audit of the 60-item judge sample (30 per benchmark); agreement (96.7% and 90.0%) and κ (0.93 and 0.80) added to Section 5.
Sun 25 Oct, Wed 28 Oct	Quyền (leader), all	Submit report and slides; presentation with live demo and backup video.
Week 45 (2–8 Nov)	Quyền, Phong	HR walkthrough with findings. Design and freeze a new benchmark of unseen, harder questions (scenario, multi-hop, conditional and temporal, paraphrased), reviewed by members.
Week 46 (9–15 Nov)	Dũng, Quyền	Run the current system and its ablations on the new benchmark; add the dense-retrieval ablation; failure analysis.
Week 47 (16–22 Nov)	Phong, Dũng	Entry-point attack tests; permission and fault checks.
Week 48 (23–29 Nov)	All	Reproduction from the recorded runs, demo hardening, final report.
Week 49 (30 Nov–2 Dec)	All	Tagged release and Round 1 presentation on 2 December.

Table 6: Weekly plan to the first final presentation, with assigned owners.

reaching 57–61% strict success with the full 4B system. The remaining tasks for the final report are the unseen harder benchmark (W45–W46) and red-team attack evaluation (W47).

8 Member Contributions and AI-Use Statement

Member	Contrib.	Role in the proposal	Completed work and repository evidence
Lê Trí Quyền (Group Leader)	33%	Retrieval and serving: ingestion, BM25 + dense index, RRF, reranking, local deployment	Corpus ingestion and normalisation (`src/ingestion/`), BM25 + TF-IDF and vector indexing with RRF fusion (`src/indexing/`), cross-encoder reranker integration (`src/retrieval/`), `llama.cpp`/Prism local serving and cross-platform Docker deployment (commits `555919a`, `8fed7c5`, `15fd823`, `ad0881b`, `bdb5d69`, `415efda`).
Võ Phạm Tuấn Dũng (Technical Lead)	33%	Agent, architecture and evaluation: system design, model selection, control loop, verifier and evaluation	Idea proposal and verification, system architecture and local model selection, codebase organisation and refactoring; bounded ReAct control loop and config-driven pipeline (`src/agent/`, `src/pipeline/`), L1 citation verifier and bounded repair (`src/verifier/`), 100-task benchmarks (`evaluation/benchmarks/`), evaluation harness, GPU/Mac runs, 60-item judge audit, statistical analysis and D2 report/slides (all remaining commits in the repository).
Nguyễn Hoàng Phong	33%	System and security: tool layer, UI, audit log, persistent store, fault and injection harness	Tool registry and code-enforced permissions (`src/tools/`), regex guardrails and PII filtering (`src/security/`), Web and Streamlit interfaces with trusted outbox (`frontend/`, `src/api/`, `src/ui/`), SQLite store and audit ledger (`src/memory/`), and fault/security test suites (commits `6dc1dbc`, `78bb069`, `2c1f3d0`, `d117349`, `7ce7e45`, `1272dc7`, `e0b4a05`, `edc9bfa`).

Table 7: Member contributions (equal 33% split) matched to proposal responsibilities, repository modules and commit history.

AI-use statement. The group used a range of AI models from Anthropic (Claude), Google (Gemini, Gemma) and OpenAI (Codex) for discussion and execution support. The problem framing, system scope, design decisions, task assignment to coding agents, supervision and code review were done by the group; every change was reviewed and approved by a group member before it was committed. AI coding agents implemented code from specifications written or approved by the group. AI models were also used, under the group's direction, to draft and cross-check benchmark questions (all questions were then reviewed by a group member), to act as the evaluation judge (an LLM estimate, checked against the group's human audit), and to draft analyses and report text that the group edited.

Verified by the group. Võ Phạm Tuấn Dũng reviewed all 100 benchmark questions against the source PDFs (23 October), audited the 60-item blind judge sample (25 October: 96.7% agreement, $\kappa = 0.93$ on HCMUT; 90.0%, $\kappa = 0.80$ on labour law), verified the architecture and code across all commits, and checked the Mac demo runs. Lê Trí Quyền verified corpus extraction, BM25/TF-IDF indexing and local `llama.cpp`/reranker serving. Nguyễn Hoàng Phong verified tool permission boundaries, guardrails and Web UI states. Automated tests use real components and no mocks (425 passed, 2 skipped, 1 failed needing an unstarted Ollama backend).

References

[1] Edoardo Debenedetti, Jie Zhang, Mislav Balunović, Luca Beurer-Kellner, Marc Fischer, and Florian Tramèr. AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2024.

[2] DeepSeek. Your first API call: Model names and legacy alias mapping, 2026. URL https://api-docs.deepseek.com/. Accessed 22 October 2026; documents V4.1-Flash-1017 serving behind the legacy Flash alias.

[3] Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. Enabling large language models to generate text with citations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

[4] Kai Greshake, Sahar Abdelnabi, Shailesh Mishra, Christoph Endres, Thorsten Holz, and Mario Fritz. Not what you've

signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection. In *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security (AISec)*, 2023.

[5] Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. Large language models cannot self-correct reasoning yet. In *International Conference on Learning Representations (ICLR)*, 2024.

[6] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[7] Varun Magesh, Faiz Surani, Matthew Dahl, Mirac Suzgun, Christopher D. Manning, and Daniel E. Ho. Hallucination-free? assessing the reliability of leading AI legal research tools. *Journal of Empirical Legal Studies*, 22(2):216–242, 2025.

[8] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*, 2023.

[9] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2023. URL https://arxiv.org/abs/2306.05685.

Appendix: Generated Result Tables

The tables below are generated automatically from the recorded runs; a script only replaces internal names by plain labels, and every number is unchanged. They are the source of the numbers in Section 5. The run, commit and checksum of each table are stored in machine-readable headers of the table files in the repository. **A.** Controller scaling comparison (Bonsai 2B vs Bonsai 4B) across both benchmarks: strict success with paired differences, then behaviour match, Recall@5, answer rate and execution-error rate. **B.** Result tables of the two runs with the 4B controller, one per benchmark.

Benchmark	Configuration	Bonsai 2B Succ. (%)	Bonsai 4B Succ. (%)	Controller effect 4B−2B (pp)
HCMUT regulations	no verifier	31.3 [22–41]	70.0 [61–79]	+38.7 [+27.3,+50.0]*
	verifier	24.0 [15–33]	64.0 [53–75]	+40.0 [+28.7,+51.3]*
	verifier + reranker	28.7 [20–37]	60.7 [50–71]	+32.0 [+21.3,+42.0]*
Labour law	no verifier	31.3 [22–41]	62.0 [52–72]	+30.7 [+20.7,+40.7]*
	verifier	28.0 [19–38]	52.7 [43–63]	+24.7 [+12.0,+37.3]*
	verifier + reranker	28.0 [19–37]	56.7 [47–67]	+28.7 [+18.0,+39.3]*

Table 8: Strict success across controller sizes (Bonsai 2B vs Bonsai 4B) on both benchmarks (LLM-judge estimate, **not human labels**; lower bound over all planned trials, errors and trials the judge could not rate count as failures; seeds averaged per question). Controller effect = 4B minus 2B (paired bootstrap differences over questions, 5000 draws). *: the 95% interval excludes 0 (descriptive, no multiple-comparison correction). Same questions in every column.

Benchmark	Configuration	Behaviour match (2B / 4B, %)	Δ controller (4B−2B)	Recall@5 (2B / 4B)	Δ controller (4B−2B)	Answer rate (2B / 4B, %)	Δ controller (4B−2B)	Execution-error rate (2B / 4B, %)	Δ controller (4B−2B)
HCMUT regulations	no verifier	58.0 / 88.0	+30.0*	0.856 / 0.989	+0.133*	55.3 / 80.7	+25.3*	0.0 / 2.0	+2.0
	verifier	45.3 / 82.7	+37.3*	0.859 / 0.989	+0.130*	41.3 / 72.7	+31.3*	0.0 / 2.0	+2.0
	verifier + reranker	46.7 / 80.0	+33.3*	0.844 / 0.978	+0.133*	42.0 / 72.7	+30.7*	0.7 / 2.0	+1.3
Labour law	no verifier	78.7 / 94.0	+15.3*	0.967 / 0.989	+0.022	74.0 / 88.0	+14.0*	0.0 / 0.0	+0.0
	verifier	63.3 / 81.3	+18.0*	0.989 / 0.989	+0.000	56.0 / 74.0	+18.0*	0.0 / 2.7	+2.7*
	verifier + reranker	65.3 / 79.3	+14.0*	0.993 / 0.993	+0.000	57.3 / 72.0	+14.7*	0.7 / 2.7	+2.0

Table 9: Metrics that need no judge across controller sizes (Bonsai 2B vs Bonsai 4B): behaviour match (expected answer → answered, expected refusal → refused), Recall@5 of gold documents in the agent's results (answerable questions; a retrieval measure on document ids, not answer correctness), answer rate and execution-error rate; percentages except Recall@5. Each cell lists 2B / 4B; Δ lists the controller effect (4B minus 2B) in percentage points (Recall@5 absolute). *: the paired 95% bootstrap interval excludes 0 (descriptive).

Benchmark	Configuration	Trials	Succ. (%)	Beh. (%)	Faith. (%)	Recall@5	MRR	Cit. P	Cit. R	Cit. F1
HCMUT regulations	no verifier	150	70.0 [61–79]†4	84.2	89.8	0.989	0.953	0.844	0.807	0.820
	verifier	150	64.0 [53–75]†6	81.2	90.3	0.989	0.952	0.726	0.693	0.704
	verifier + reranker	150	60.7 [50–71]†10	77.1	89.3	0.978	0.937	0.733	0.704	0.714

Table 10: Results with the Bonsai 4B controller on the HCMUT regulations benchmark, three configurations (450 trials; random seeds 42, 123, 2026). *Succ.* is strict success (appropriate behaviour, correctness and completeness both full, every claim supported): a lower bound over all trials, with a 95% bootstrap interval over questions; execution errors count as failures, and †k marks k trials the judge could not rate, also counted as failures. Succ., Beh. (behaviour match) and Faith. (faithfulness) are estimates by the LLM judge of Section 5, **not human labels**. Judge–human agreement on strict success over 30 human-labelled items: 97% agreement, $\kappa = 0.93$ (1 reviewer(s)). Recall@5 and MRR measure whether the gold documents appear in the agent's search and read results (answerable questions; all cells are 0.98–0.99, so they cannot separate the configurations). Cit. P/R/F1 compare the cited document ids with the gold ids (nothing cited counts as 0); they are measured on document ids and say nothing about answer correctness. Red entries are not measured.

Benchmark	Configuration	Ans/Ref/Clar/Err	Err. (%)	Trial p50/p95 (s)	Call p50/p95 (s)	tok/s	Tok. in/out	Calls/trial	Unknown	Judge USD
HCMUT regulations	no verifier	121/23/3/3	2.0	*n/r*	*n/r*	*n/r*	10233/666	3.13	4	0.270
	verifier	109/35/3/3	2.0	*n/r*	*n/r*	*n/r*	12923/807	3.47	6	0.282
	verifier + reranker	109/37/1/3	2.0	*n/r*	*n/r*	*n/r*	12070/763	3.37	10	0.312

Table 11: Operating measurements and judge cost with the Bonsai 4B controller on the HCMUT regulations benchmark. Ans/Ref/Clar/Err are the numbers of answered, refused, clarified and execution-error trials; Err. is the execution-error rate. n/r (not reported): latency was recorded while several workers ran at once, so it is not reportable. Call = one controller model call; tok/s = generated tokens per second. Tokens are controller input/output per trial (trials with complete usage only). Judge cost is the usage cost reported by the API gateway over all retained attempts (1.064 USD for all configurations; not an invoice).

Comparison	Scope	Tasks	Δ Succ. (pp)	Δ Beh. match (pp)	Δ Rec.@5 (pp)	Δ Cit. F1 (pp)	Δ Err. (pp)
verifier vs no verifier	HCMUT regulations	50	-6.0 [-16.0,+4.0]	-5.3 [-14.7,+3.3]	+0.0 [+0.0,+0.0]	-11.6 [-22.0,-1.7]	+0.0 [-3.3,+2.7]
verifier + reranker vs verifier	HCMUT regulations	50	-3.3 [-13.3,+6.7]	-2.7 [-9.3,+3.3]	-1.1 [-3.3,+0.0]	+1.0 [-6.2,+7.7]	+0.0 [-2.7,+2.7]
verifier + reranker vs no verifier	HCMUT regulations	50	-9.3 [-18.7,+0.0]	-8.0 [-16.7,+0.0]	-1.1 [-3.3,+0.0]	-10.6 [-20.7,-0.5]	+0.0 [-4.0,+3.3]

Table 12: Differences between configurations on the HCMUT regulations benchmark (treatment minus baseline, percentage points) with 95% paired bootstrap intervals over questions (seeds averaged per question; 5000 resamples). Succ. is strict success (LLM-judge lower bound), Beh. the behaviour match, Rec.@5 the Recall@5 of gold documents, Cit. F1 the citation F1 on document ids, Err. the execution-error rate. There is no multiple-comparison correction; an interval that includes 0 shows no detectable difference at this sample size.

Benchmark	Row	Trials	Succ. (all, %)	Succ. (answerable, %)	Answer rate (%)	Cit. P	Cit. R
	closed-book (diagnostic)	50/50	2.0 [0–6]	0.0 [0–0]	2.2	–	–
	oracle (diagnostic)	45/45	82.2 [71–93]	82.2 [71–93]	100.0	1.000	1.000
HCMUT regulations	no verifier (reference)	135	–	69.6 [60–79]	88.1	0.844	0.807
	verifier (reference)	135	–	61.5 [50–73]	80.7	0.726	0.693
	verifier + reranker (reference)	135	–	59.3 [49–70]	79.3	0.733	0.704

Table 13: Two diagnostic baselines on the HCMUT regulations benchmark: one seed (42), the same 4B controller, sampler, domain prompt and constrained decoding, *no tools*, verifier off. They are not system configurations and are never pooled with them. *Closed-book*: the question only, all questions. *Oracle*: the full gold records are given up front in the format of a read result, answerable questions only (refusal questions have no gold context and are excluded). Succ. (all) is over every question the row ran; Succ. (answerable) is restricted to answerable questions (the like-for-like column), with a bootstrap interval over questions; both are LLM-judge lower bounds and errors count as failures. Answer rate = answered / all recorded trials (execution errors count as not answered) on answerable questions. Cit. P/R compare cited ids with gold ids (meaningful for the oracle and the system rows; a closed-book model has no document ids). Reference rows repeat the answerable-only results of the three configurations (seeds averaged). The oracle bounds what the 4B controller achieves with perfect retrieval; closed-book bounds what the model's own knowledge achieves.

Benchmark	Configuration	Trials	Succ. (%)	Beh. (%)	Faith. (%)	Recall@5	MRR	Cit. P	Cit. R	Cit. F1
	no verifier	150	62.0 [52–72]$^{\dagger 7}$	91.6	87.3	0.989	0.952	0.886	0.830	0.841
Labour law	verifier	150	52.7 [43–63]$^{\dagger 6}$	75.7	86.3	0.989	0.952	0.759	0.719	0.728
	verifier + reranker	150	56.7 [47–67]$^{\dagger 4}$	76.7	85.6	0.993	0.959	0.726	0.678	0.691

Table 14: Results with the Bonsai 4B controller on the Labour law benchmark, three configurations (450 trials; random seeds 42, 123, 2026). *Succ.* is strict success (appropriate behaviour, correctness and completeness both full, every claim supported): a lower bound over all trials, with a 95% bootstrap interval over questions; execution errors count as failures, and $\dagger k$ marks k trials the judge could not rate, also counted as failures. Succ., Beh. (behaviour match) and Faith. (faithfulness) are estimates by the LLM judge of Section 5, **not human labels**. Judge–human agreement on strict success over 30 human-labelled items: 90% agreement, $\kappa = 0.80$ (1 reviewer(s)). Recall@5 and MRR measure whether the gold documents appear in the agent's search and read results (answerable questions; all cells are 0.99–0.99, so they cannot separate the configurations). Cit. P/R/F1 compare the cited document ids with the gold ids (nothing cited counts as 0); they are measured on document ids and say nothing about answer correctness. Red entries are not measured.

Benchmark	Configuration	Ans/Ref/Clar/Err	Err. (%)	Trial p50/p95 (s)	Call p50/p95 (s)	tok/s	Tok. in/out	Calls/trial	Unknown	Judge USD
	no verifier	132/14/4/0	0.0	*n/r*	*n/r*	*n/r*	9070/686	3.11	7	0.360
Labour law	verifier	111/31/4/4	2.7	*n/r*	*n/r*	*n/r*	11382/840	3.65	6	0.327
	verifier + reranker	108/37/1/4	2.7	*n/r*	*n/r*	*n/r*	11477/865	3.69	4	0.269

Table 15: Operating measurements and judge cost with the Bonsai 4B controller on the Labour law benchmark. Ans/Ref/Clar/Err are the numbers of answered, refused, clarified and execution-error trials; Err. is the execution-error rate. n/r (not reported): latency was recorded while several workers ran at once, so it is not reportable. Call = one controller model call; tok/s = generated tokens per second. Tokens are controller input/output per trial (trials with complete usage only). Judge cost is the usage cost reported by the API gateway over all retained attempts (1.103 USD for all configurations; not an invoice).

Comparison	Scope	Tasks	Δ Succ. (pp)	Δ Beh. match (pp)	Δ Rec.@5 (pp)	Δ Cit. F1 (pp)	Δ Err. (pp)
verifier vs no verifier	Labour law	50	-9.3 [-18.7,+0.0]	-12.7 [-20.7,-5.3]	+0.0 [+0.0,+0.0]	-11.2 [-19.8,-3.5]	+2.7 [+0.7,+5.3]
verifier + reranker vs verifier	Labour law	50	+4.0 [-6.0,+14.0]	-2.0 [-8.7,+5.3]	+0.4 [-2.2,+3.3]	-3.7 [-10.6,+3.2]	+0.0 [-2.7,+2.7]
verifier + reranker vs no verifier	Labour law	50	-5.3 [-15.3,+4.0]	-14.7 [-24.0,-6.0]	+0.4 [-2.2,+3.3]	-14.9 [-25.3,-5.7]	+2.7 [+0.0,+6.0]

Table 16: Differences between configurations on the Labour law benchmark (treatment minus baseline, percentage points) with 95% paired bootstrap intervals over questions (seeds averaged per question; 5000 resamples). Succ. is strict success (LLM-judge lower bound), Beh. the behaviour match, Rec.@5 the Recall@5 of gold documents, Cit. F1 the citation F1 on document ids, Err. the execution-error rate. There is no multiple-comparison correction; an interval that includes 0 shows no detectable difference at this sample size.

Benchmark	Row	Trials	Succ. (all, %)	Succ. (answerable, %)	Answer rate (%)	Cit. P	Cit. R
	closed-book (diagnostic)	50/50	24.0 [12–36]	24.4 [13–38]	2.2	–	–
	oracle (diagnostic)	45/45	75.6 [62–87]	75.6 [62–87]	97.8	0.967	0.944
Labour law	no verifier (reference)	135	–	60.7 [50–71]	95.6	0.886	0.830
	verifier (reference)	135	–	51.1 [40–62]	80.7	0.759	0.719
	verifier + reranker (reference)	135	–	54.1 [43–64]	78.5	0.726	0.678

Table 17: Two diagnostic baselines on the Labour law benchmark: one seed (42), the same 4B controller, sampler, domain prompt and constrained decoding, *no tools*, verifier off. They are not system configurations and are never pooled with them. *Closed-book*: the question only, all questions. *Oracle*: the full gold records are given up front in the format of a read result, answerable questions only (refusal questions have no gold context and are excluded). Succ. (all) is over every question the row ran; Succ. (answerable) is restricted to answerable questions (the like-for-like column), with a bootstrap interval over questions; both are LLM-judge lower bounds and errors count as failures. Answer rate = answered / all recorded trials (execution errors count as not answered) on answerable questions. Cit. P/R compare cited ids with gold ids (meaningful for the oracle and the system rows; a closed-book model has no document ids). Reference rows repeat the answerable-only results of the three configurations (seeds averaged). The oracle bounds what the 4B controller achieves with perfect retrieval; closed-book bounds what the model's own knowledge achieves.

Benchmark	Configuration	Answer expected: A/R/C/Err	Refusal expected: A/R/C/Err	Over-refusal (%)	Unsafe answer (%)	Refusal P (%)	Refusal R (%)	Strict match (%)
	no verifier	119/10/3/3	2/13/0/0	9.8	13.3	56.5	86.7	88.0
HCMUT regulations	verifier	109/20/3/3	0/15/0/0	17.4	0.0	42.9	100.0	82.7
	verifier + reranker	107/24/1/3	2/13/0/0	18.9	13.3	35.1	86.7	80.0

Table 18: Behaviour against what the benchmark expects on the HCMUT regulations benchmark (no judge). The rows "answer expected" and "refusal expected" give the numbers of answered / refused / clarified / error trials (A/R/C/Err). Over-refusal = refused or clarified although an answer exists; unsafe answer = answered although the question should be refused; refusal P/R treat *refused* as the positive class (non-error trials); strict match counts errors as mismatches.

Benchmark	Configuration	Closed-book guessable (%)	Not guessable (%)	Refusal tasks (%)
	no verifier	73.3 [62–84] (n=15)	67.8 [53–81] (n=30)	73.3 [47–100] (n=5)
HCMUT regulations	verifier	80.0 [64–93] (n=15)	52.2 [37–67] (n=30)	86.7 [60–100] (n=5)
	verifier + reranker	71.1 [56–84] (n=15)	53.3 [40–68] (n=30)	73.3 [33–100] (n=5)

Table 19: Strict success split by whether the question can be guessed on the HCMUT regulations benchmark (LLM-judge lower bound, errors count as failures). "Closed-book guessable": a strong model answered the question correctly *without* the corpus (itself an LLM estimate), so these questions measure the model's own knowledge as much as grounding. "Not guessable": partially correct or unsolved without the corpus. Refusal questions have no such split. Intervals are bootstrap intervals over questions; n = questions in the group.

Benchmark	Configuration	Answer expected: A/R/C/Err	Refusal expected: A/R/C/Err	Over-refusal (%)	Unsafe answer (%)	Refusal P (%)	Refusal R (%)	Strict match (%)
	no verifier	129/2/4/0	3/12/0/0	4.4	20.0	85.7	80.0	94.0
Labour law	verifier	109/18/4/4	2/13/0/0	16.8	13.3	41.9	86.7	81.3
	verifier + reranker	106/24/1/4	2/13/0/0	19.1	13.3	35.1	86.7	79.3

Table 20: Behaviour against what the benchmark expects on the Labour law benchmark (no judge). The rows "answer expected" and "refusal expected" give the numbers of answered / refused / clarified / error trials (A/R/C/Err). Over-refusal = refused or clarified although an answer exists; unsafe answer = answered although the question should be refused; refusal P/R treat *refused* as the positive class (non-error trials); strict match counts errors as mismatches.

Benchmark	Configuration	Closed-book guessable (%)	Not guessable (%)	Refusal tasks (%)
	no verifier	52.8 [31–72] (n=12)	63.6 [51–76] (n=33)	73.3 [53–93] (n=5)
Labour law	verifier	38.9 [19–58] (n=12)	55.6 [42–68] (n=33)	66.7 [40–93] (n=5)
	verifier + reranker	52.8 [33–72] (n=12)	54.5 [41–68] (n=33)	80.0 [67–93] (n=5)

Table 21: Strict success split by whether the question can be guessed on the Labour law benchmark (LLM-judge lower bound, errors count as failures). "Closed-book guessable": a strong model answered the question correctly *without* the corpus (itself an LLM estimate), so these questions measure the model's own knowledge as much as grounding. "Not guessable": partially correct or unsolved without the corpus. Refusal questions have no such split. Intervals are bootstrap intervals over questions; n = questions in the group.